

DEVIXO SOLUTIONS | DATA SCIENCE INTERNSHIP

Task 04

E-Commerce Business Analytics

From raw data to business decisions: cleaning, exploratory analysis, KPIs, interactive dashboard, insights, recommendations and machine learning.

Item	Detail
Project	E-Commerce Business Analytics
Dataset	12,220 raw order lines (Jan 2023 - Dec 2024), 20 columns
Tools	Python (Pandas, NumPy, Matplotlib, Seaborn, scikit-learn), SQL (SQLite), Excel, Plotly interactive dashboard
Deliverables	Cleaned dataset, 25 charts, 6-page dashboard, Excel report, SQL queries, insights and recommendations
Prepared by	Haris Khursheed

1. Project Overview

This project turns a raw, messy e-commerce order dataset into a complete decision-making solution for a business. It follows the five parts of the task brief and adds a machine learning bonus.

Task part	What was done	Where in this report
Part 1	Data cleaning, outlier and missing-value treatment, feature engineering	Section 3
Part 2	Exploratory data analysis with 21 visualisations	Section 5
Part 3	Key performance indicators (revenue, profit, AOV, growth, retention)	Section 4
Part 4	Interactive 6-page dashboard with slicers	Section 6
Part 5	20 insights and 12 actionable recommendations	Sections 7 and 8
Bonus	Segmentation, churn prediction, sales and demand forecasting	Section 9

Business problem

The business has strong revenue but a thin overall profit margin, and revenue declined in 2024. Management needs to know which products, customers, discounts and regions create profit, where money is being lost, and what to do about it.

Headline results

- Total revenue **\$2,758,380**, total profit **\$307,043**, overall margin **11.1%**.
- Electronics gives 62% of revenue but only a 5% margin; Fashion and Beauty earn about 29% margin.
- Discounts above 10% destroy profit and do not improve customer ratings.
- 2024 revenue fell **19.5%** versus 2023 while margin stayed flat, so the problem is customer volume, not pricing.
- The top 20% of customers generate 66% of revenue.

2. How to Run the Project

Everything runs from one command. Python 3.9 or newer is needed.

Step by step

1. Unzip the project folder **ecommerce_analytics** and open a terminal inside it.
2. (Recommended) create and activate a virtual environment.

```
python -m venv venv
venv\Scripts\activate      # Windows
source venv/bin/activate   # Mac / Linux
```

3. Install the libraries.

```
pip install -r requirements.txt
```

4. Run the whole pipeline (about 10 seconds).

```
python run_all.py
```

5. Double-click **dashboard.html** to open the interactive dashboard (works offline). Read **outputs/insights_and_recommendations.md** and open **outputs/Business_Analytics_Report.xlsx** for the full results.

Project files

File	Purpose
01_generate_raw_data.py	Creates the raw, messy dataset (skip this if you use your own CSV)
02_data_cleaning.py	Part 1: cleaning and feature engineering
03_eda_kpis.py	Parts 2 and 3: 21 charts and the KPI tables
04_ml_models.py	Bonus: RFM segmentation, churn, forecasts
05_build_dashboard.py	Part 4: builds dashboard.html
06_insights_report.py	Part 5: insights and recommendations
07_sql_and_excel.py	Runs the 15 SQL queries, builds the Excel workbook
run_all.py	Runs all steps in order

To use a different dataset, place it at **data/raw_ecommerce_orders.csv**, skip step 01, and rename its columns to match those used in 02_data_cleaning.py.

3. Part 1: Data Preparation

The raw file contained 12,220 rows and 20 columns with realistic quality problems. After cleaning, the dataset has **12,000 rows and 35 columns, 0 missing values and 0 duplicate orders**.

Issue	Found	Treatment
Duplicates	220 duplicate rows	Removed (exact and by Order ID)
Mixed date formats	4 different text formats	Parsed into proper dates
Symbols in numbers	"\$" in Sales, "%" in Discount	Stripped and converted to numeric
Inconsistent text	Upper case, spaces, "N. America", "Europ"	Trimmed and standardised
Unit price outliers	20 values above 3x product median	Replaced with product median
Quantity outliers	25 values (e.g. -3, 50, 150)	Re-derived from Sales / price
Age outliers	15 values (0, 5, 120, 999)	Set to missing, then imputed
Missing Rating	1,501 missing (12%)	Product median plus a flag column
Missing Age, Gender, Channel, Payment	607 / 353 / 385 / 200	Same customer's other orders, else median / mode
Missing Profit	108 missing	Sales x category average margin
High-value orders	1,341 IQR outliers	Kept: genuine big-ticket items such as laptops

Engineered features

Year, Month, Quarter, YearMonth, Day of Week, Is Weekend, Shipping Days, Profit Margin %, Is Loss, Age Group, Discount Band and Order Value Band were added to support the analysis.

4. Part 3: Key Performance Indicators

KPI	Value
Total Revenue	\$2,758,380
Total Profit	\$307,043
Total Orders	12,000
Average Order Value	\$229.87
Profit Margin	11.13%
Customer Count	2,205
Revenue Growth (2024 vs 2023)	-19.5%
Profit Growth (2024 vs 2023)	-18.9%
Revenue per Customer	\$1,250.97
Repeat Customer Rate	79.6%
Customers inactive > 180 days	48.3%
Average Rating	3.95
Loss-making Orders	15.6%

Profit and revenue both fell by about 19% in 2024, so margin held steady at about 11%. The decline comes from fewer orders, not from pricing. Almost half of the customers have not bought in the last 180 days.

5. Part 2: Exploratory Data Analysis

Twenty-one visualisations cover trends, top and poor performers, relationships between variables, regional performance and customer behaviour. The most important ones are shown here; all 21 are in the charts folder.

5.1 Trends and seasonality

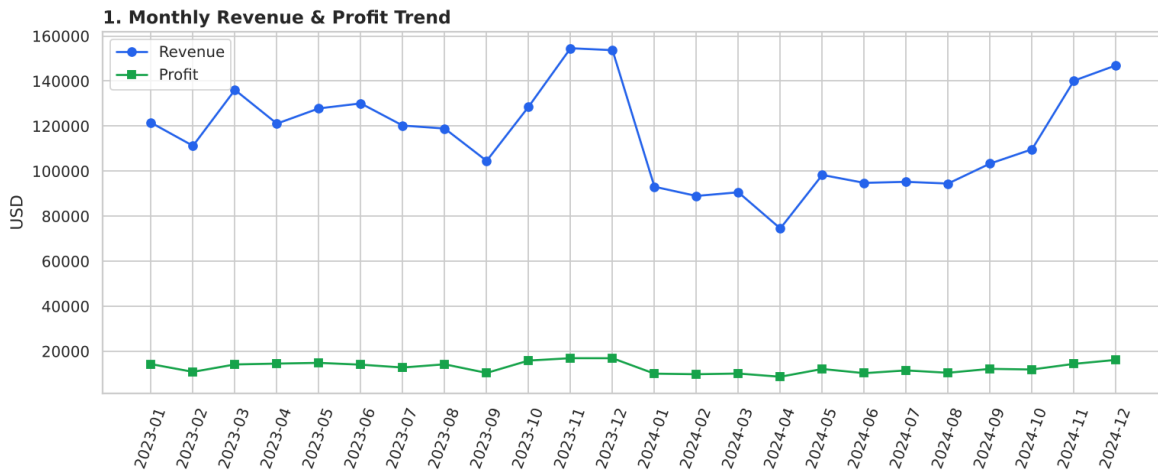


Figure 1. Monthly revenue and profit. Revenue peaks in Nov-Dec and dropped sharply from January 2024.

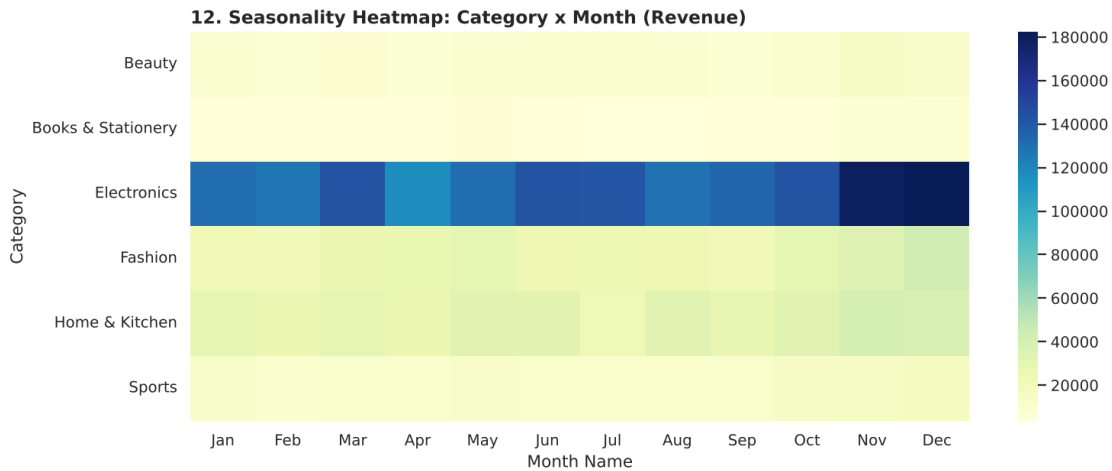


Figure 2. Seasonality heat-map by category and month: Q4 drives demand, led by Electronics.

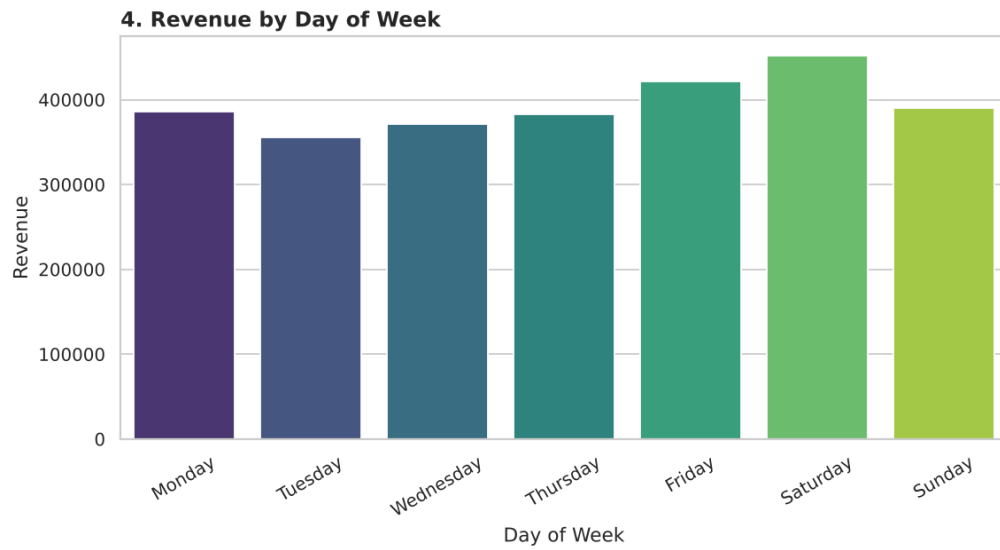


Figure 3. Revenue by day of week: Saturday is strongest, Tuesday weakest.

5.2 Categories and products

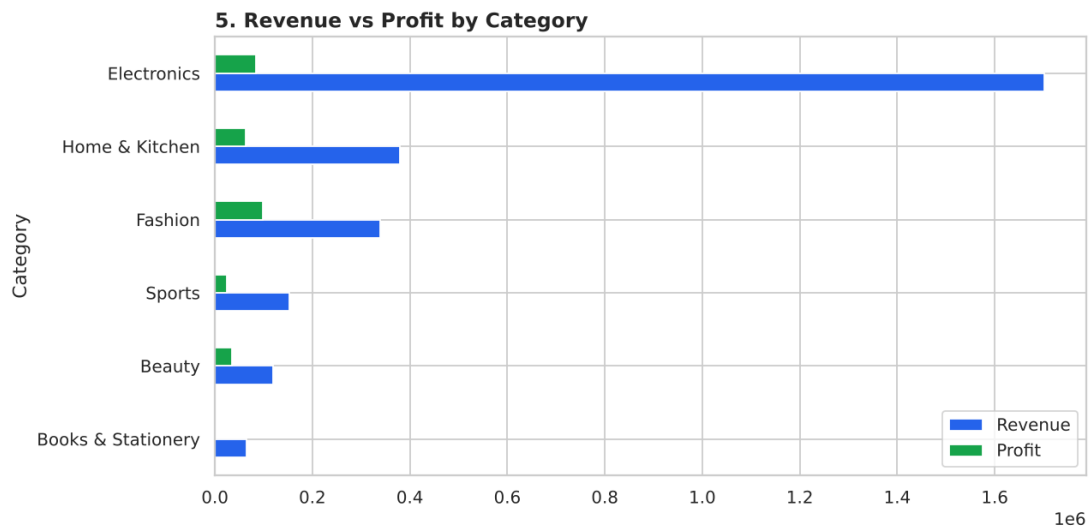


Figure 4. Revenue versus profit by category. Electronics is large but earns little profit.

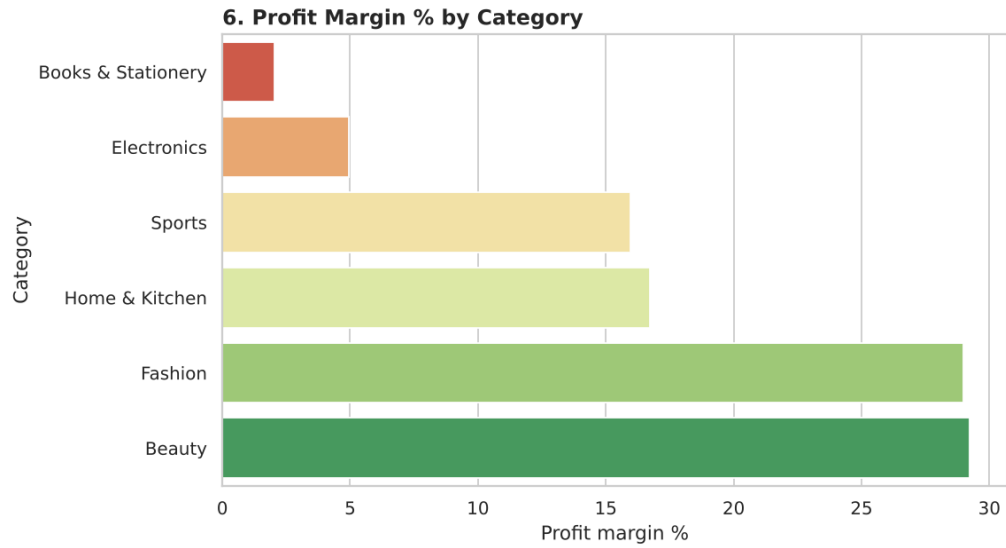


Figure 5. Profit margin by category: Beauty and Fashion lead, Books & Stationery is near zero.

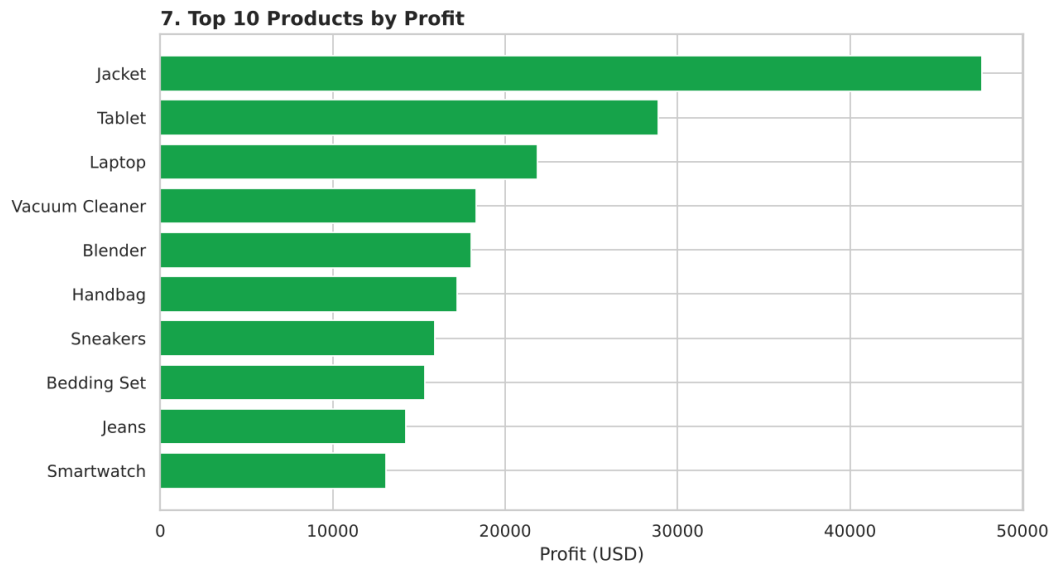


Figure 6. Top 10 products by profit.

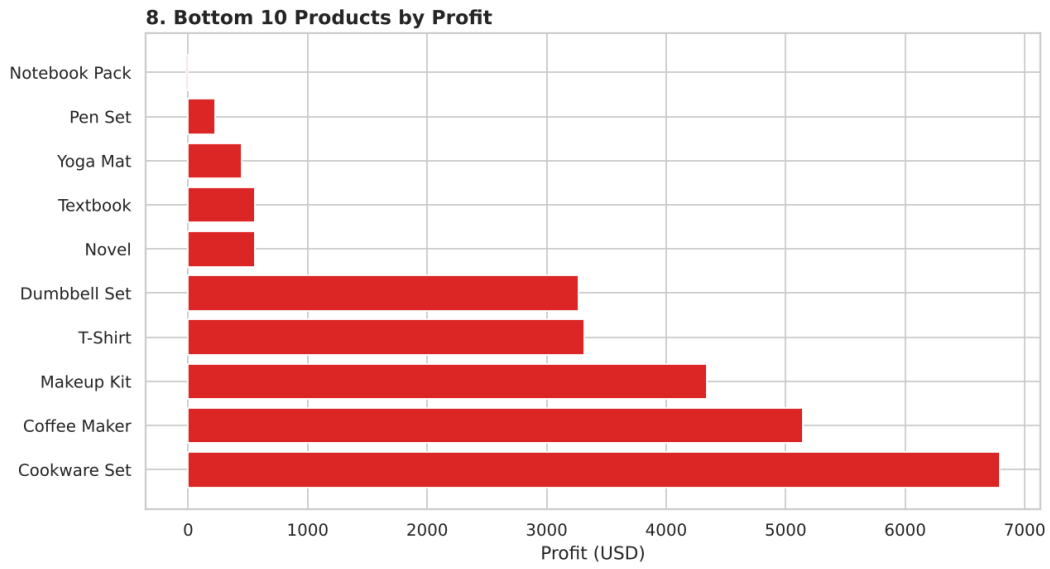


Figure 7. Bottom 10 products by profit: low-priced Books & Stationery items.

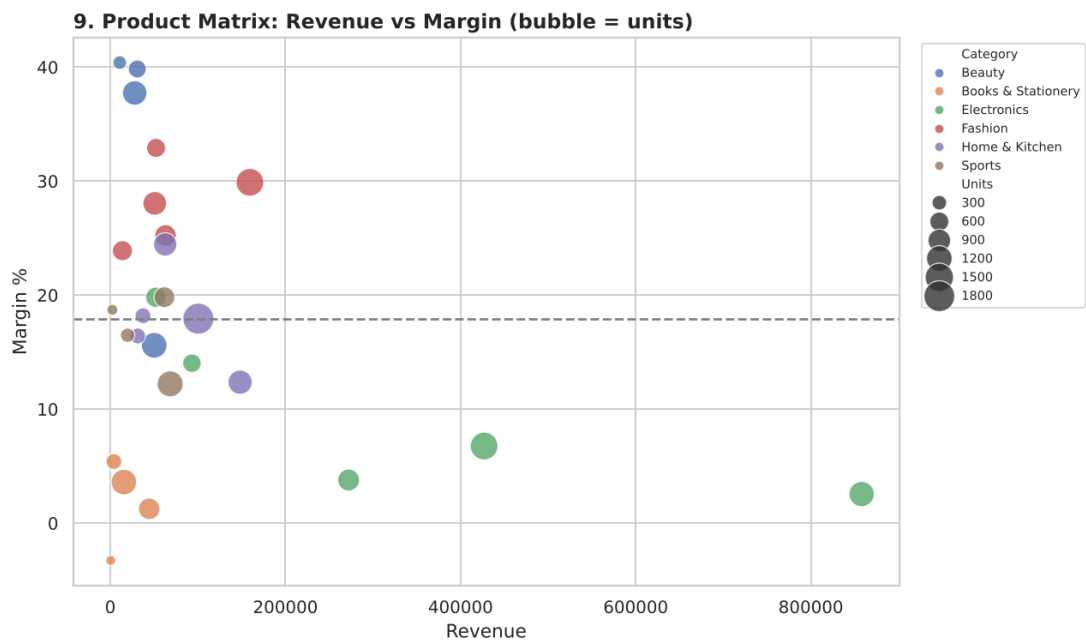


Figure 8. Product matrix of revenue against margin (bubble size = units sold).

5.3 Relationships

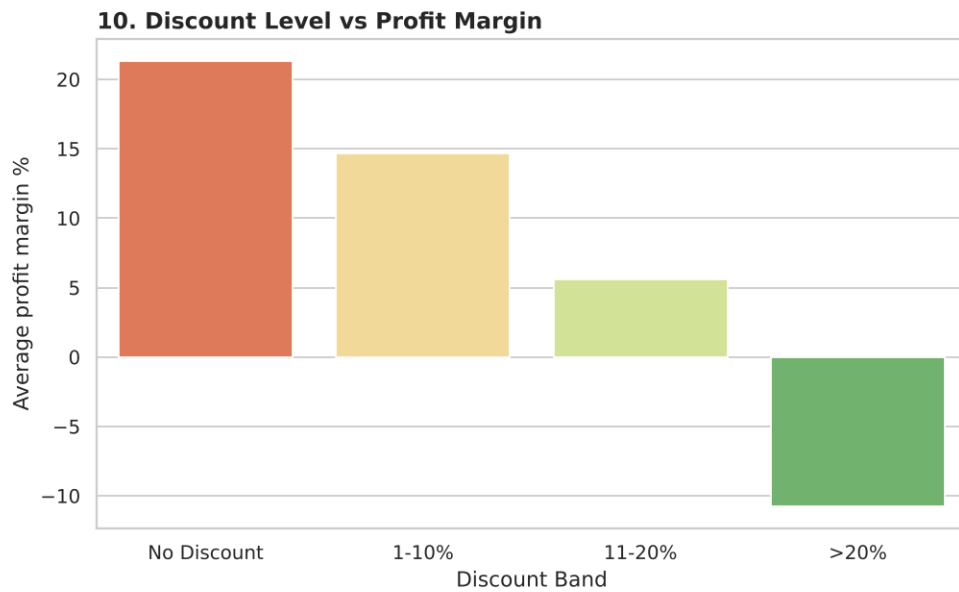


Figure 9. Average margin by discount level: margin turns negative above 20% discount.

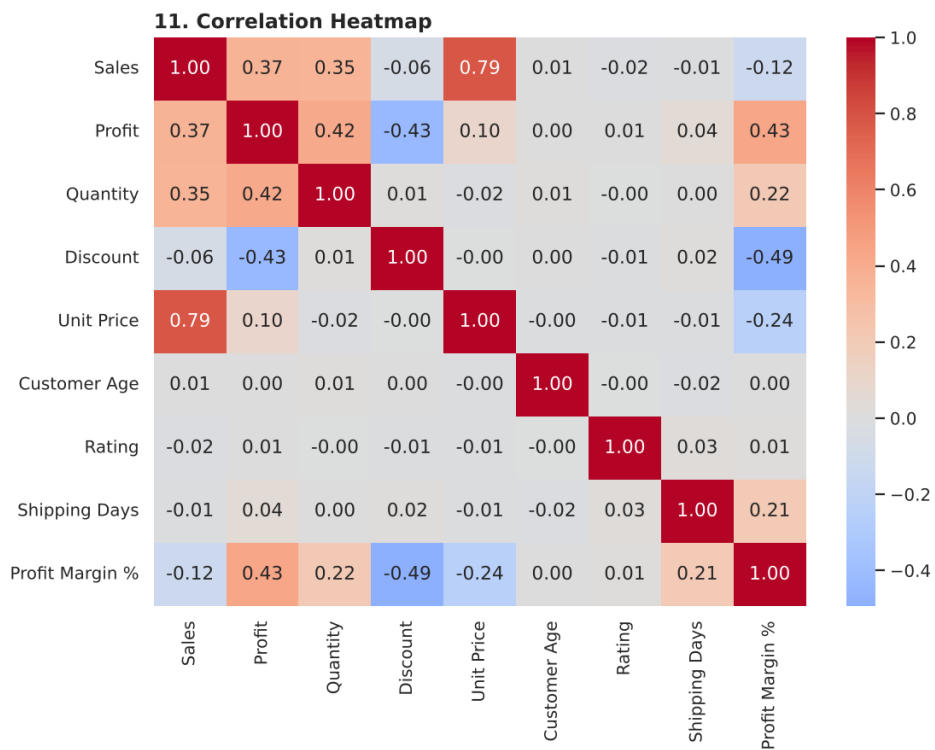
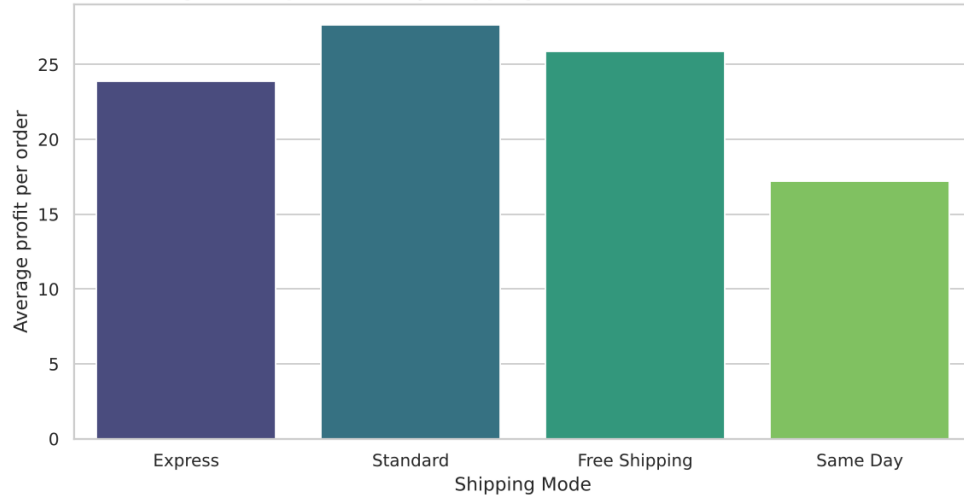
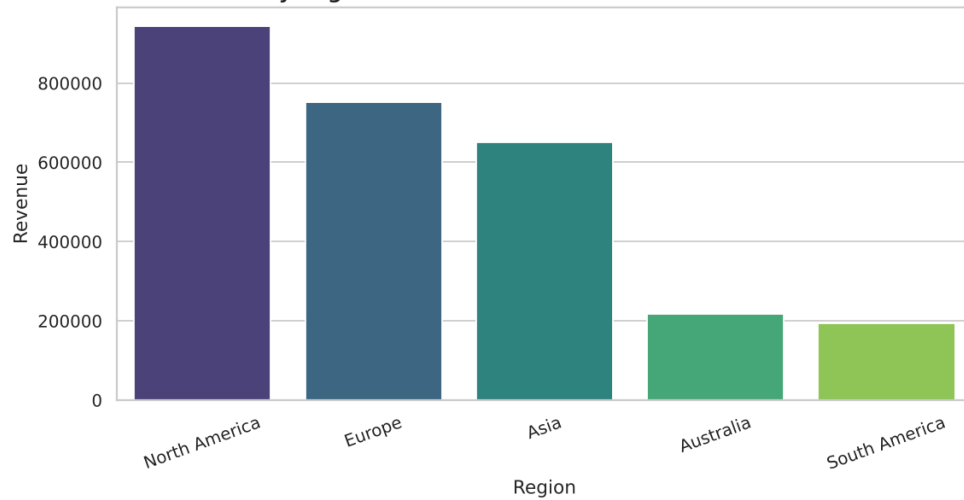


Figure 10. Correlation heat-map of the numeric variables.

20. Average Profit per Order by Shipping Mode*Figure 11. Average profit per order by shipping mode: Same Day is the weakest.*

5.4 Regions, channels and customers

13. Revenue by Region*Figure 12. Revenue by region.*

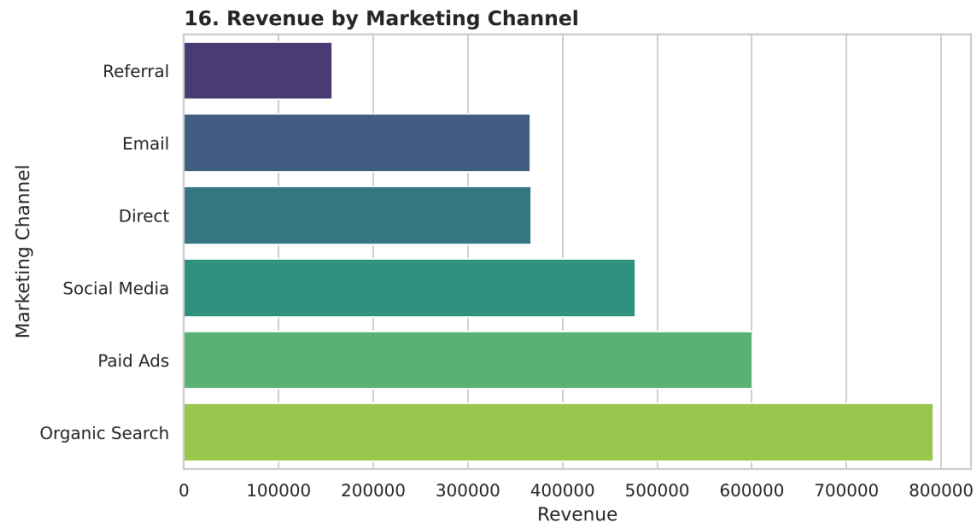


Figure 13. Revenue by marketing channel.

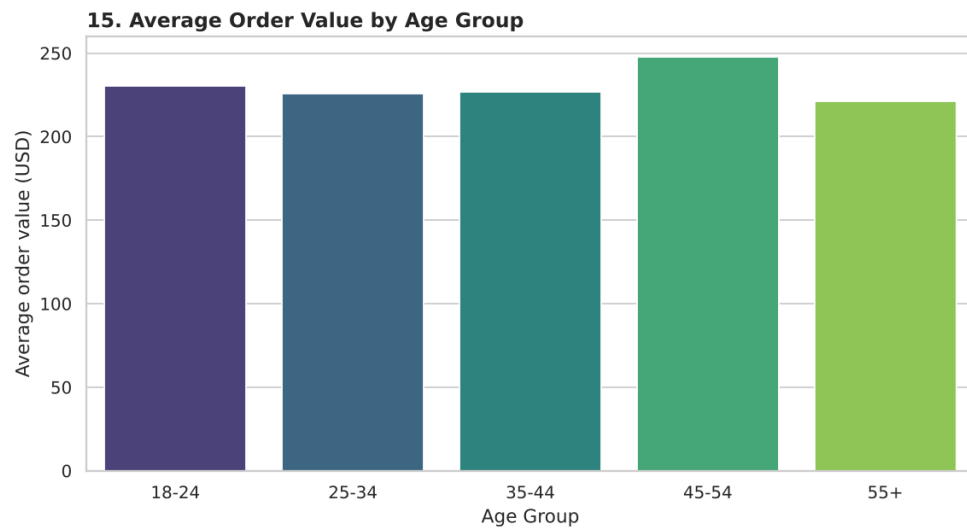


Figure 14. Average order value by age group.

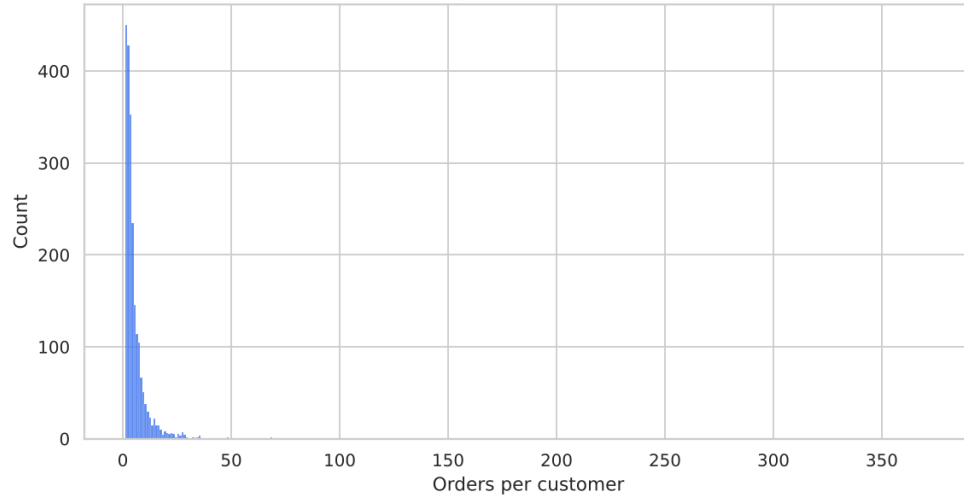
18. Customer Purchase Frequency Distribution

Figure 15. Customer purchase frequency distribution.

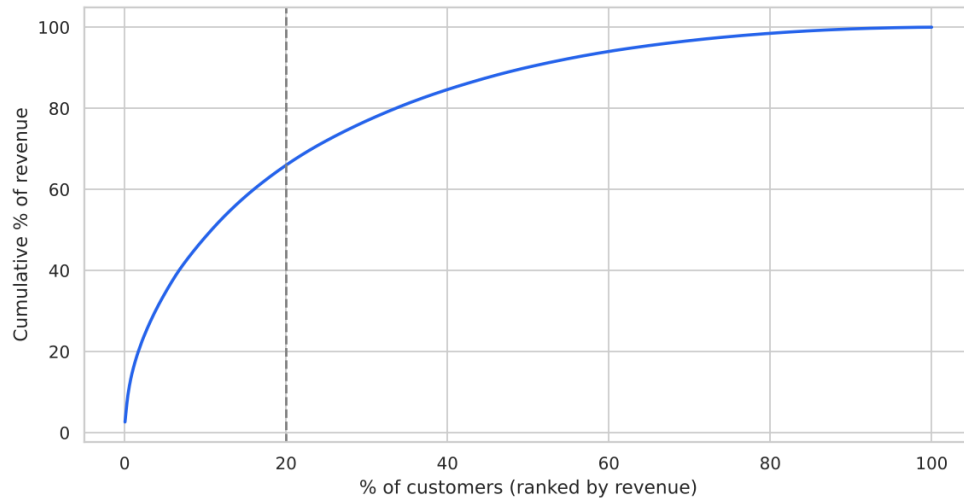
19. Pareto: Customer Revenue Concentration

Figure 16. Pareto curve: the top 20% of customers generate 66% of revenue.

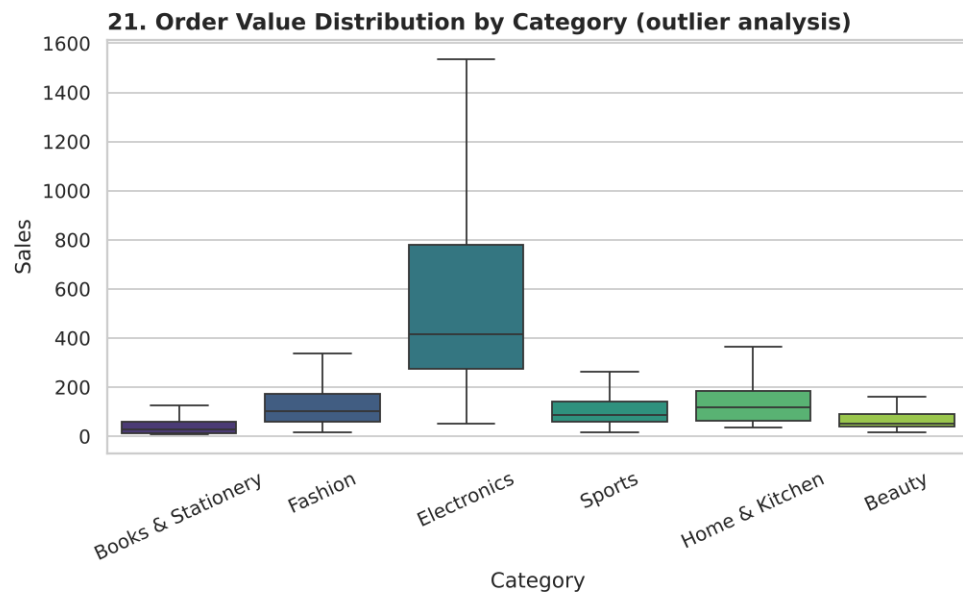


Figure 17. Order value distribution by category (outlier analysis).

6. Part 4: Interactive Dashboard

The dashboard is one self-contained file, **dashboard.html**, with six pages and about 30 interactive charts. Four slicers (Year, Region, Category, Customer Segment) at the top filter every KPI and chart on every page. Charts support hover tooltips, zoom and image download. The screenshots below were taken from the running dashboard.

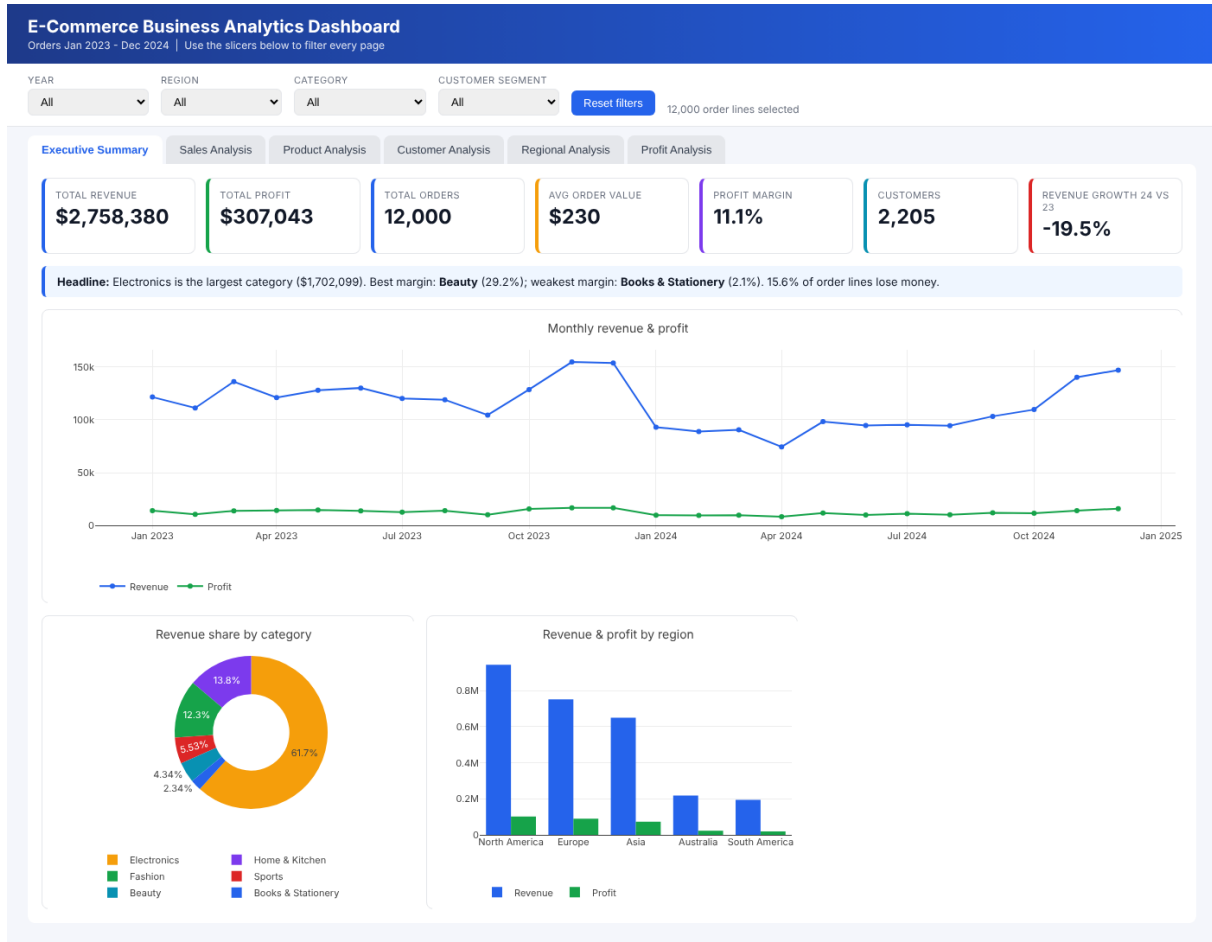


Figure 18. Page 1: Executive Summary. KPI cards, automatic headline, monthly trend, category share and regional comparison.

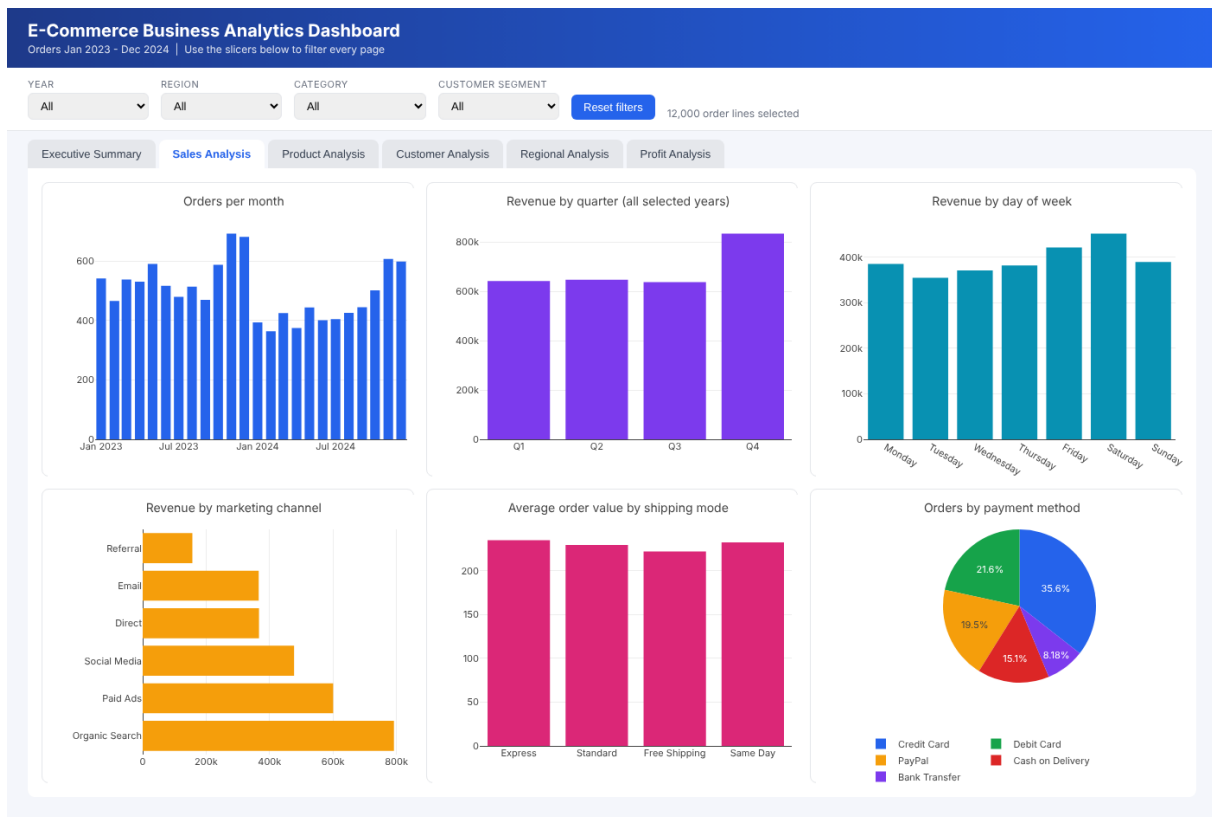


Figure 19. Page 2: Sales Analysis. Orders per month, quarter, day of week, marketing channel, shipping mode and payment method.

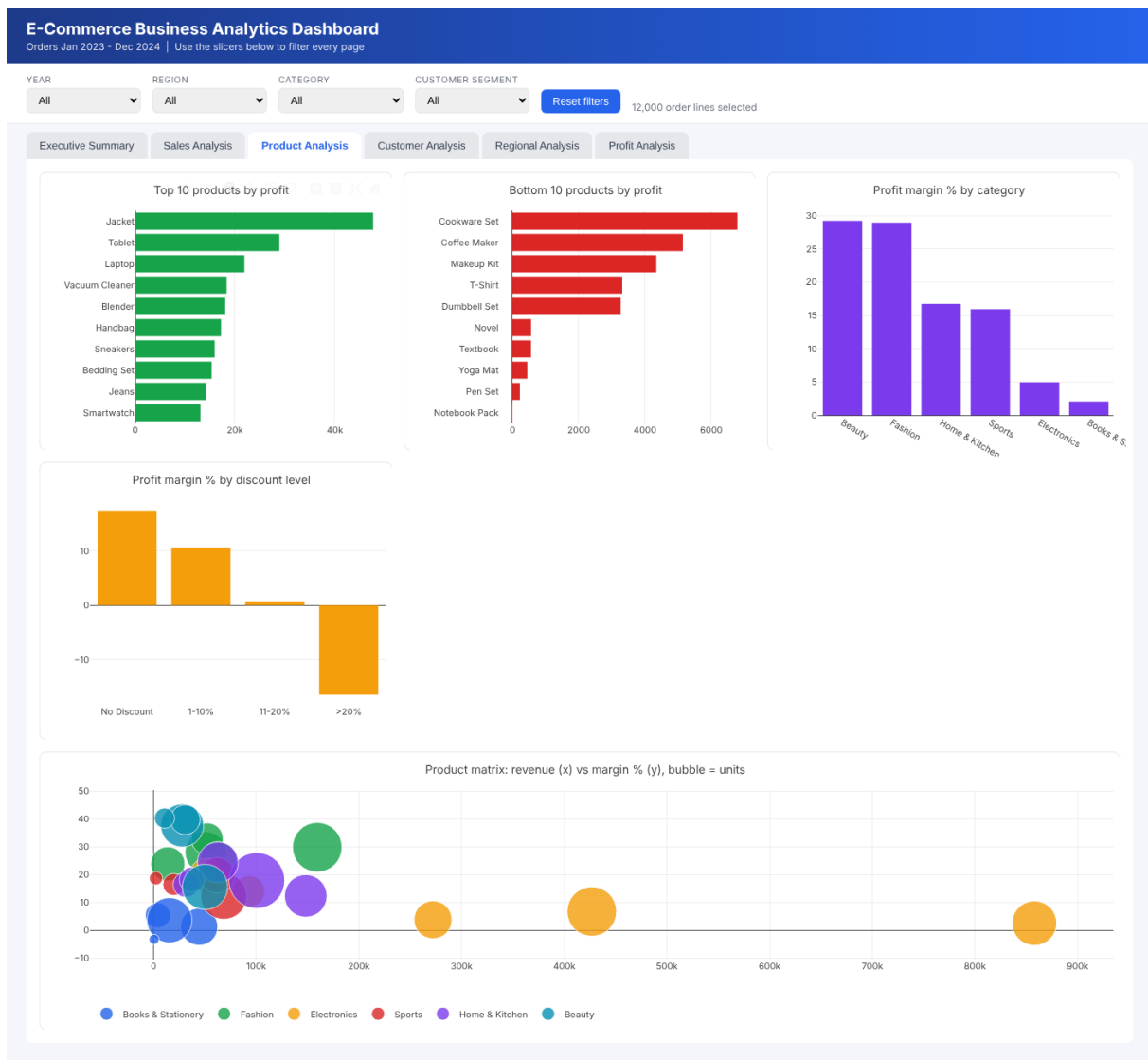


Figure 20. Page 3: Product Analysis. Top and bottom products, margin by category and discount level, product matrix.

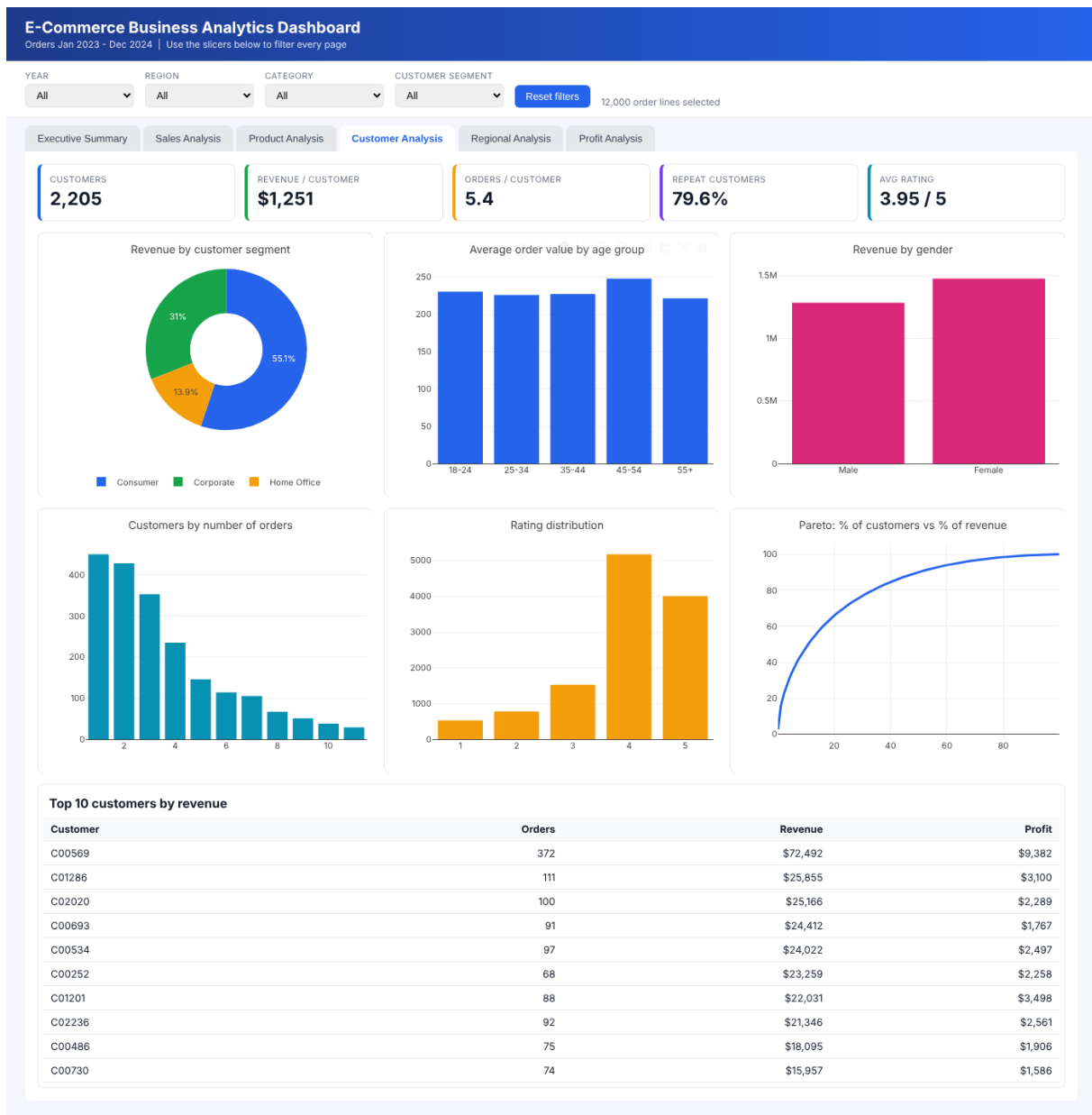


Figure 21. Page 4: Customer Analysis. Segments, age, gender, purchase frequency, ratings, Pareto curve and top-10 customers.

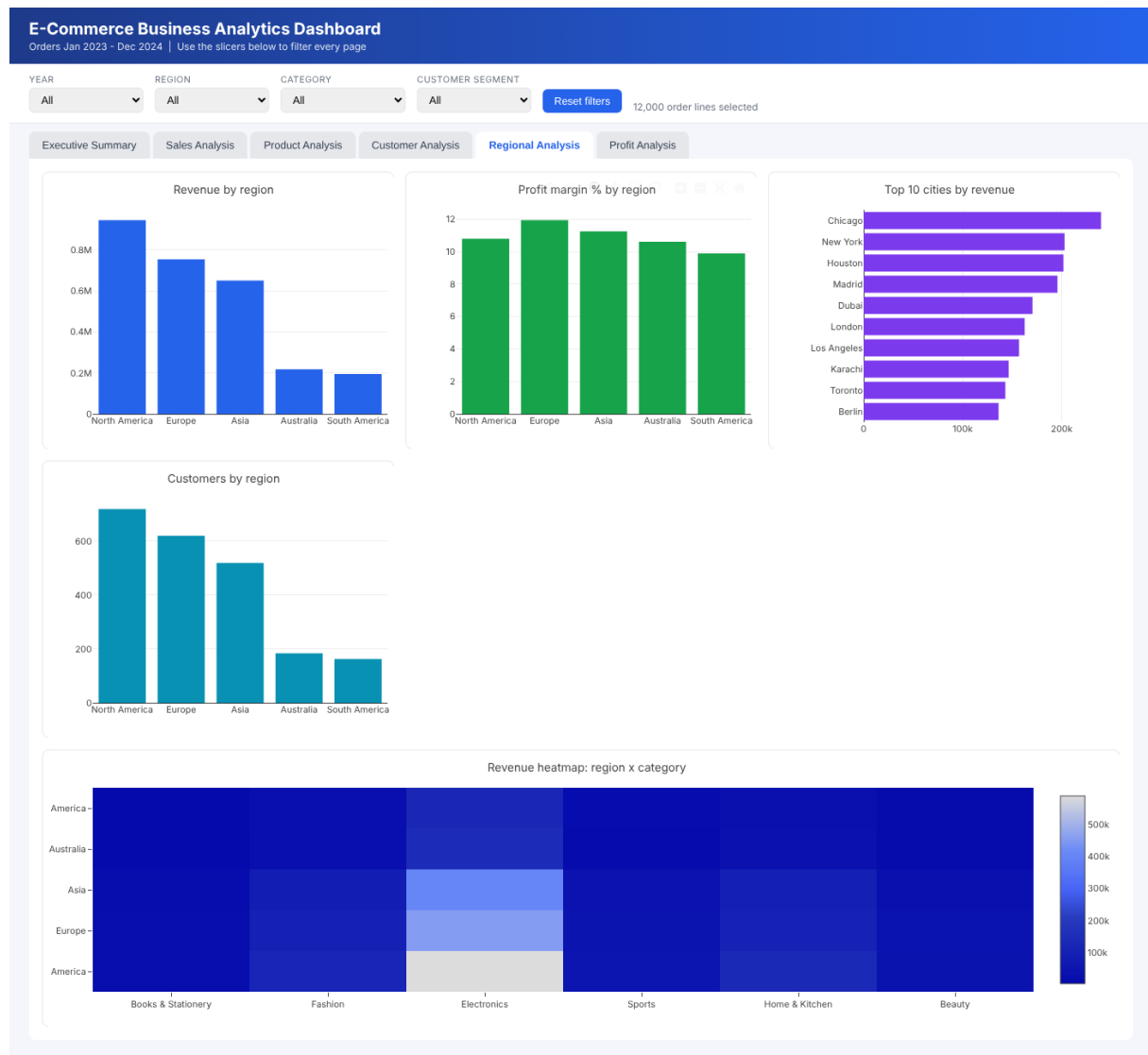


Figure 22. Page 5: Regional Analysis. Revenue, margin, top cities, customers and a region by category heat-map.

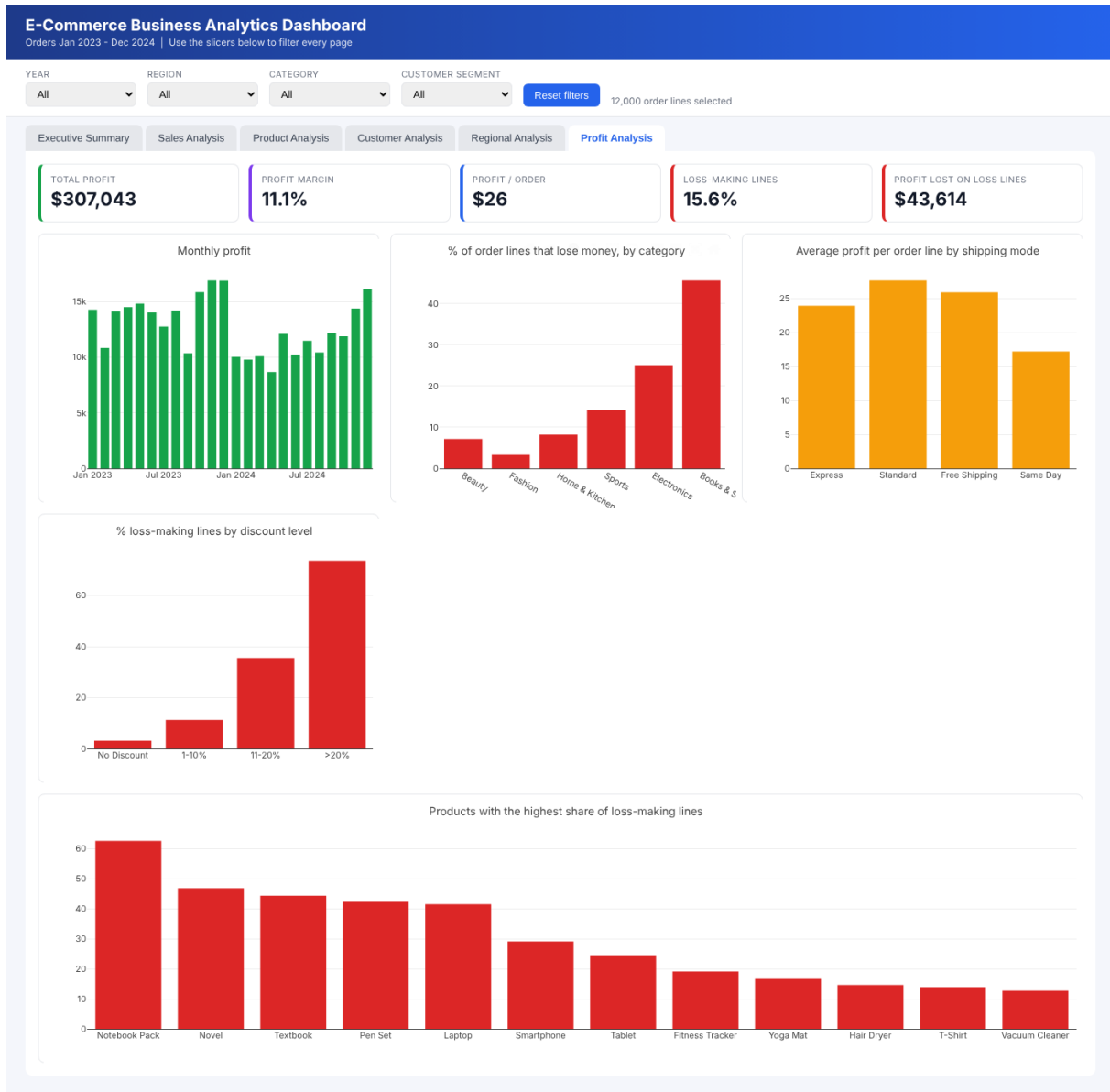


Figure 23. Page 6: Profit Analysis. Monthly profit, loss-making share by category, shipping mode, discount level and product.

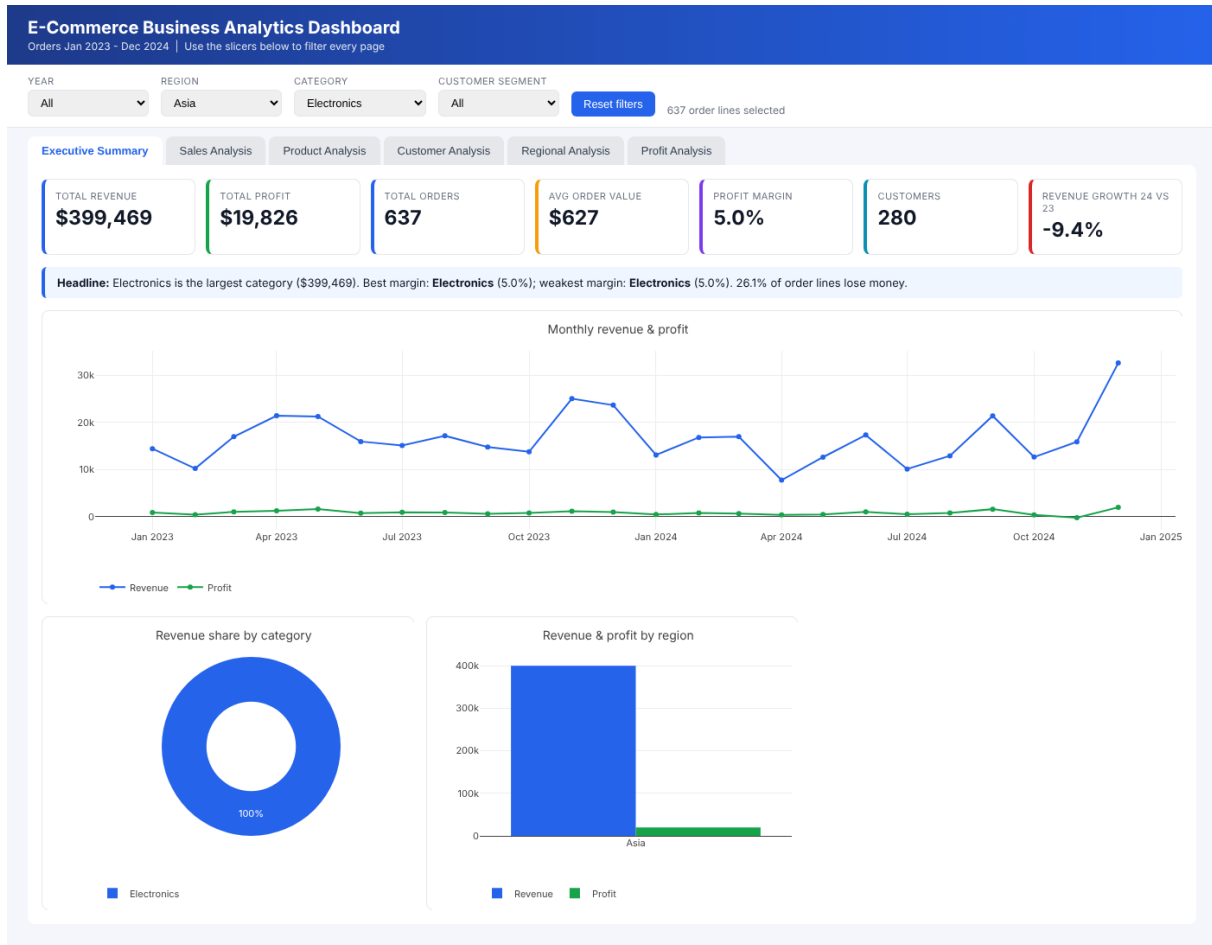


Figure 24. Slicers in action: Region = Asia and Category = Electronics. Every KPI and chart recalculates instantly.

7. Part 5: Key Insights

Every figure below is computed directly from the cleaned data.

1. **Business size & health.** 12,000 orders from 2,205 customers produced \$2,758,380 revenue and \$307,043 profit - a thin overall margin of 11.1%.
2. **Revenue is concentrated in one category.** Electronics brings \$1,702,099 (62% of revenue) but only 5.0% margin and 28% of total profit.
3. **Highest-margin categories are small.** Beauty (29.2%) and Fashion (29.0%) earn 6x the margin of Electronics; Fashion alone delivers 32% of profit from 12% of revenue.
4. **Books & Stationery barely earns money.** Margin is 2.1% and 46% of its order lines lose money.
5. **Discounts above 10% destroy profit.** Margin falls from 17.4% (no discount) to 0.7% (11-20% off) and -16.4% (>20% off). 74% of orders with >20% discount lose money.
6. **Discounts do not buy happier customers.** Average rating is almost identical with no discount (3.96) and with >20% off (3.94), so deep discounts are not improving satisfaction.
7. **Laptops drive revenue but not profit.** Laptops generate \$857,892 (31% of all revenue) at a 2.6% margin, with 41% of laptop orders loss-making.
8. **Loss-making orders are a major leak.** 15.6% of order lines have negative profit, costing \$43,614 in total - 14% of the profit the company kept.
9. **Same-day shipping is unprofitable.** Same Day margin is 7.4% vs 12.0% for Standard, and 34% of Same Day orders lose money, while AOV is no higher (\$232 vs \$229).
10. **Strong seasonality.** Dec and Nov are the two best months; Q4 delivers 30% of annual revenue versus an even 25%.
11. **2024 revenue fell 19.5% vs 2023** (\$1,528,344 -> \$1,230,036) while the margin stayed flat (11.1% - > 11.2%) - the decline is a volume/customer problem, not a pricing problem.
12. **Customer revenue is highly concentrated.** The top 20% of customers generate 66% of revenue; the 275 'Champions' (RFM) alone produce 49% of revenue.
13. **A large group of customers has already gone.** 784 customers (36%) are in the 'Lost / Churned' RFM segment (long inactive, ~1-2 orders), who together account for just 5.5% of revenue but represent a shrinking active base.
14. **Repeat purchasing is healthy.** 80% of customers ordered more than once, so retention (not acquisition) is the lever that protects revenue.
15. **Regions are similar in margin; North America is the largest.** North America brings 34% of revenue; margins range only from 9.9% (South America) to 11.9% (Europe), so regional differences come from scale, not efficiency.
16. **Marketing channels perform similarly.** Organic Search is the biggest channel (29% of revenue); Referral has the best margin (11.9%). Paid Ads spends are not delivering a better margin (10.6%) than free Organic Search (11.0%).
17. **Weekends and Fridays sell most.** Saturday is the strongest day (\$452,179) and Tuesday the weakest (\$355,086), a 27% gap.
18. **Age 25-44 is the core audience.** These two age groups contribute 62% of revenue; the 55+ group is only 5%.

19. **Several products are structurally unprofitable.** Notebook Pack, Novel, Textbook, Pen Set, Laptop have more than 40% of order lines losing money.
20. **Forecast.** Seasonality-adjusted model projects \$571,018 revenue over the next 6 months (Jan-Jun 2025) vs \$540,203 in the same months of 2024 (+6%). Back-test error on the final 6 months was about 4% (see ML report).

8. Part 5: Recommendations

Each recommendation links a finding to a concrete action and an estimated impact. Impact figures are simple estimates that ignore volume loss from price changes, so they should be validated with a pilot or A/B test.

1. Cap discounts at 10% and stop >20% promotions

Finding: Orders above 10% discount have 0.7% / -16.4% margins and 74% of >20% orders lose money; ratings are unchanged by discounts.

Action: Introduce a hard 10% discount ceiling; require manager approval above that; replace deep discounts with bundles or loyalty points.

Expected impact: Re-pricing the 2,973 orders above 10% discount to 10% would add about \$69,949 profit before volume effects (a volume drop would offset part of this, so test on one category first).

2. Re-price or restructure the Laptop line

Finding: Laptops are 31% of revenue at a 2.6% margin and 41% of orders lose money.

Action: Raise laptop list prices 3-5%, remove laptops from discount campaigns, bundle with high-margin accessories (headphones, smartwatch).

Expected impact: Every 1 point of laptop margin is worth about \$8,579.

3. Shift marketing & stock toward Fashion and Beauty

Finding: Fashion (29.0%) and Beauty (29.2%) have the best margins and almost no losses.

Action: Move 15-20% of paid-ads budget from Electronics to Fashion/Beauty; feature Jackets, Handbags and Perfume in homepage and email campaigns.

Expected impact: If Fashion/Beauty revenue grows 20%, profit rises by about \$26,666.

4. Fix or exit Books & Stationery low-value lines

Finding: Margin is only 2.1% with 46% loss-making lines; Notebook Pack, Pen Set, Textbook are the weakest products.

Action: Set a minimum order value / free-shipping threshold, remove discounts on this category, cross-sell to larger baskets, and delist the permanently loss-making SKUs.

Expected impact: Removes up to \$3,215 of losses made on loss-making Books & Stationery orders over the two years.

5. Charge for Same-Day delivery

Finding: Same-Day margin is 7.4% vs 12.0% Standard and 34% of its orders lose money.

Action: Add a \$10-15 same-day fee or limit it to orders above \$150; promote Standard shipping with a delivery-date guarantee.

Expected impact: Protects roughly \$3,120 (about \$10 per loss-making same-day order).

6. Launch a win-back campaign for lapsed customers

Finding: 784 customers are in the Lost segment and 2024 revenue dropped 20%.

Action: Send a 3-step email/SMS sequence (reminder, no-discount personalised recommendations, 10% max coupon) to customers inactive over 180 days; track reactivation rate.

Expected impact: Reactivating just 10% of lost customers at the average Loyal-segment spend adds about \$51,161 (assuming half of their historic spend).

7. Protect the Champions with a VIP programme

Finding: 275 Champions produce 49% of revenue; the top 20% of customers give 66%.

Action: Create tiered loyalty rewards, early access to new products, and a dedicated support line; review these accounts monthly.

Expected impact: Losing even 10% of Champions would cost about \$134,391.

8. Use the churn model to trigger retention offers

Finding: The Random Forest churn model flags customers likely to stop buying; 100 high-value customers currently exceed 60% churn probability (file: high_value_at_risk_customers.csv).

Action: Give the list to CRM weekly; contact the highest-value at-risk customers with personal outreach and a non-discount incentive (free shipping, early access).

Expected impact: These 100 customers represent \$352,814 of past revenue.

9. Build a Q4 inventory and campaign plan

Finding: Dec and Nov are peak months and Q4 delivers 30% of revenue.

Action: Pre-order stock for top sellers by September, run margin-aware Black Friday offers (bundles, not price cuts), and add support staff in Nov-Dec.

Expected impact: Avoids stock-outs in the 3 highest-revenue months and protects margin during peak discount pressure.

10. Rebalance marketing spend toward efficient channels

Finding: Paid Ads margin (10.6%) is no better than Organic Search (11.0%); Email and Referral have the best margins (11.8% / 11.9%).

Action: Invest in SEO, email automation and a referral reward scheme; cap paid-ads CPA and track profit-per-channel instead of revenue.

Expected impact: Lower acquisition cost for the same revenue.

11. Run targeted promotions on weak days

Finding: Tuesday revenue is 27% lower than Saturday.

Action: Use flash sales (bundle-based, not price cuts) and email send-times on Tuesday-Thursday to smooth demand.

Expected impact: Lifting the three weakest days to the average would add roughly \$36,726 of extra revenue per year (about \$4,088 profit at the current margin).

12. Grow in under-penetrated regions through targeted offers

Finding: Australia and South America together give only 15% of revenue and show below-average margins (10.6% and 9.9% vs 11.1% overall).

Action: Test local-currency pricing, regional free-shipping thresholds and local payment methods before spending on large acquisition campaigns.

Expected impact: Small pilot first; scale only if margin exceeds the company average.

9. Bonus: Machine Learning

9.1 Customer segmentation (RFM + K-Means)

Customers were grouped on Recency, Frequency and Monetary value into four segments. Champions are only 275 customers but produce 49% of revenue.

Segment	Customers	Recency (days)	Orders	Avg spend	Revenue share
Champions	275	51.7	20.0	\$4,887	48.7%
Loyal	671	307.4	4.8	\$1,305	31.7%
At Risk / Occasional	475	35.6	4.2	\$814	14.0%
Lost / Churned	784	350.0	1.6	\$194	5.5%

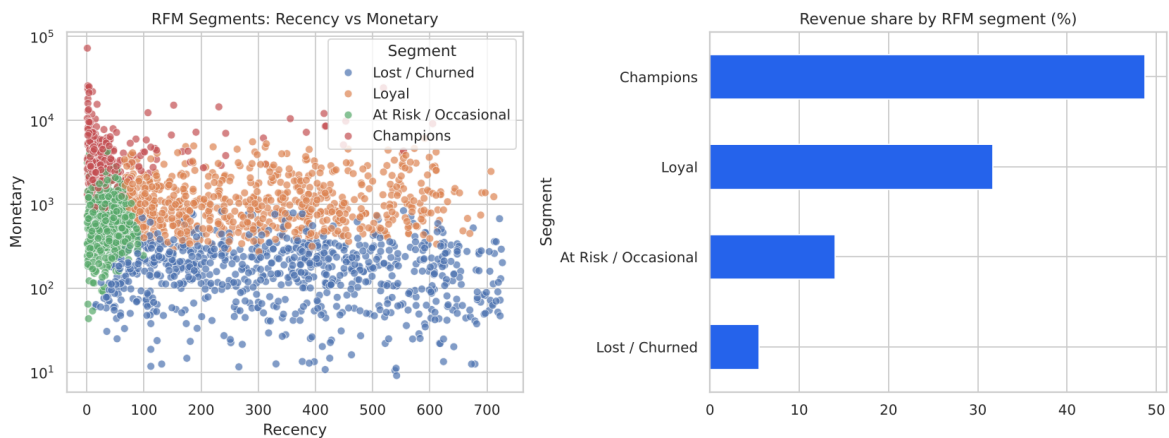


Figure 25. RFM segments and their revenue share.

9.2 Churn prediction

A Random Forest predicts whether a customer stops buying in the six months after 30 June 2024, using only information from before that date (no data leakage). It reaches **AUC 0.71 and accuracy 64%**, better than logistic regression (AUC 0.68). Recency, tenure and total spend are the main drivers. The model is moderate; the list of high-value at-risk customers is exported for the retention team.

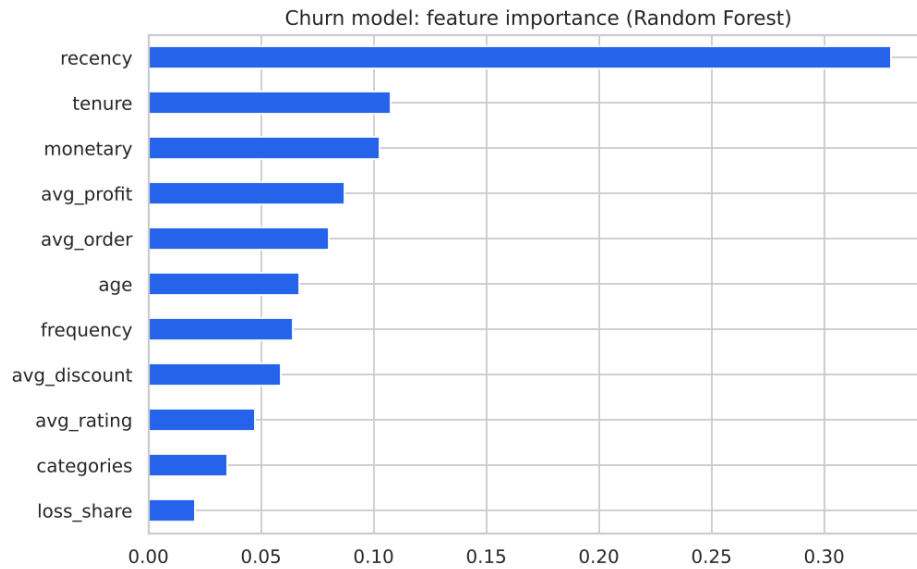


Figure 26. Churn model feature importance.

9.3 Sales forecast

A seasonal-level model (recent level multiplied by a monthly seasonal index) forecasts the next six months. Back-tested on the last six months it had a MAPE of about **4.3%**, versus 14.3% for a naive average. The projection for January to June 2025 is **\$571,018**.

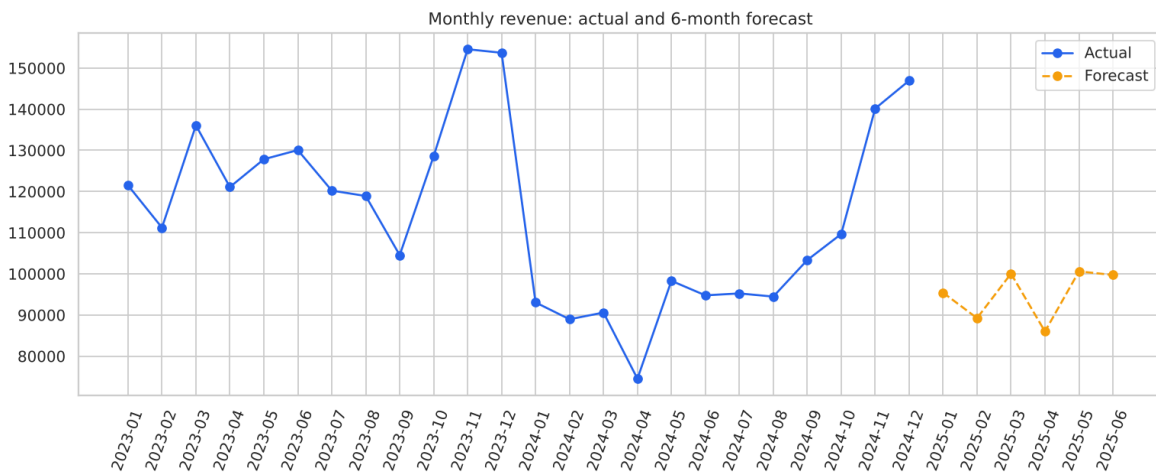


Figure 27. Monthly revenue: actual and six-month forecast.

9.4 Product demand

Next-quarter unit demand is forecast for every product and saved in outputs/product_demand_forecast.csv, to support stock planning.

10. Limitations and Conclusion

Limitations

- The dataset is synthetic, generated so the project runs anywhere. The pipeline works unchanged on a real dataset once column names match.
- The churn model is moderate (AUC 0.71) because customer behaviour in this data is largely random.
- The forecast seasonal index uses all 24 months, so the back-test is slightly optimistic; with two years of history the forecast is indicative.
- Recommendation impacts are estimates and should be validated with an A/B test.

Conclusion

The analysis shows that this business grows revenue in the wrong places: big-ticket Electronics and deep discounts bring sales but little profit, while Fashion and Beauty earn most of the profit. Capping discounts at 10%, repricing laptops, charging for same-day delivery, and winning back lapsed customers are the highest-value actions. Together with the dashboard and models, these give management a clear, data-backed plan.